



Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Pre-Processing + Ensemble Models

**MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E
INFORMÁTICA**
Inteligência Artificial para as Telecomunicações
2025/2026



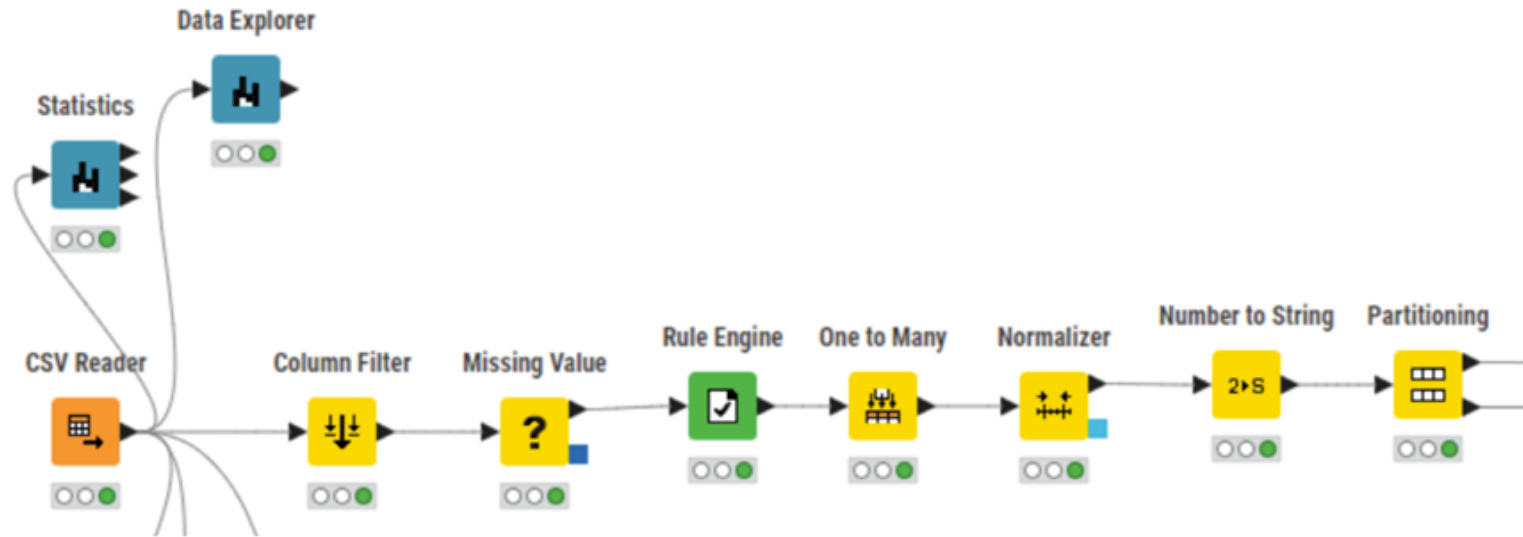
Objectives

- The primary goal of this task is to create a comprehensive and efficient workflow tailored specifically for pre-processing the dataset to develop models for **predicting survival** on the Titanic dataset .

Dataset - Features

- **12 attributes**
 - **PassengerId:** An identification number for each passenger;
 - **Survived:** This is the target variable (0 = The passenger did not survive, 1 = The passenger survived);
 - **Pclass:** Passenger class (1 = Upper class, 2 = Middle class, 3 = Lower class);
 - **Name:** Full name of the passenger;
 - **Sex:** Gender of the passenger (male or female);
 - **Age:** Age of the passenger in years (with some missing values);
 - **SibSp:** Number of siblings/spouses aboard the Titanic;
 - **Parch:** Number of parents/children aboard the Titanic;
 - **Ticket:** Ticket number;
 - **Fare:** The ticket fare the passenger paid;
 - **Cabin:** Cabin number (with many missing values);
 - **Embarked:** Port of embarkation (with C = Cherbourg, Q = Queenstown, S = Southampton).

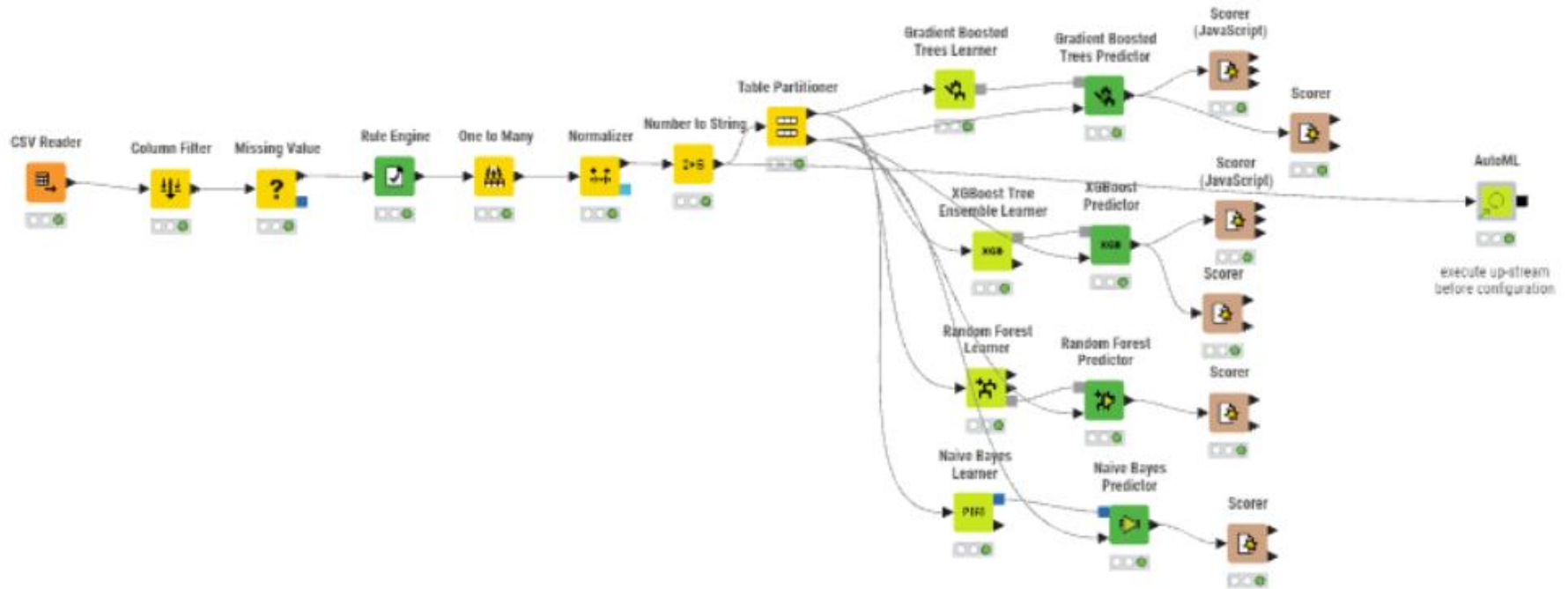
Pre-Processed Workflow



Models

- **Gradient Boost Trees**
- **XGboost**
- **Random Forest**
- **Naïve Bayes**
- **Auto ML (Naïve Bayes, Random Frorest and XGBoost)**

Final Workflow



Evaluation – Gradient Booster

Scorer View

Confusion Matrix

	0 (Predicted)	1 (Predicted)	
0 (Actual)	96	18	84.21%
1 (Actual)	21	44	67.69%
	82.05%	70.97%	

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
78.21%	21.79%	0.524	140	39

Evaluation – XGBoost

Scorer View

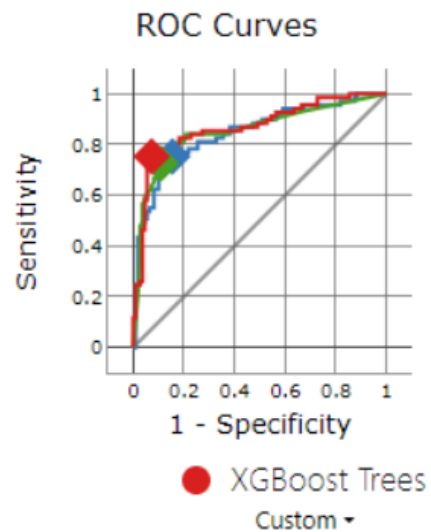
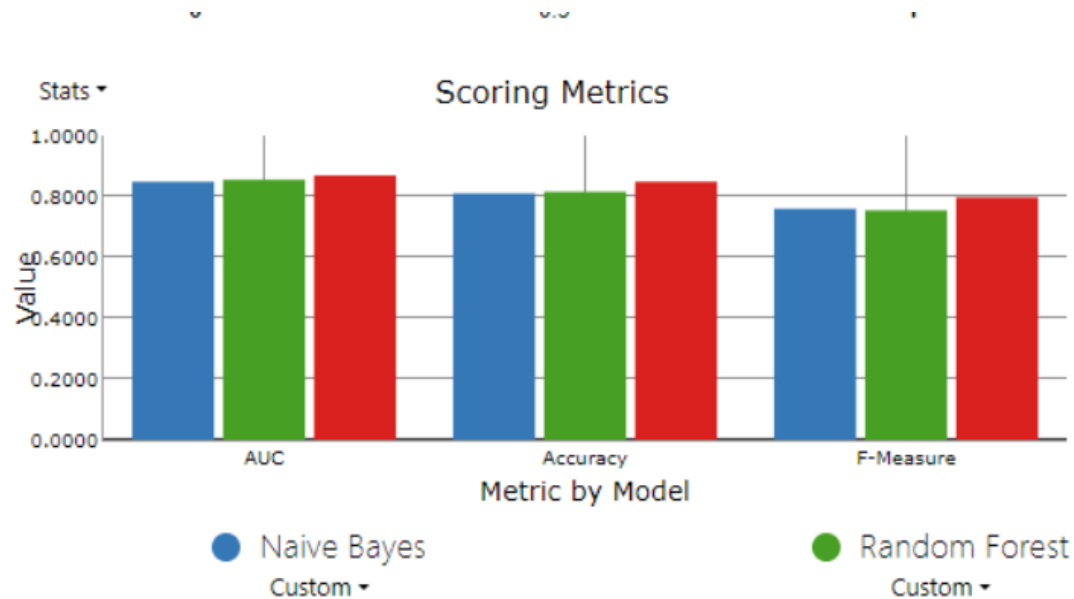
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Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
78.21%	21.79%	0.524	140	39

Evaluation – Auto ML



Evaluation

Model	Accuracy	Cohen's Kappa	Error (%)
Gradient Boosted Trees	78.21%	0.52	21.79%
XGBoost	78.21%	0.52	21.79%
Random Forest	76.54%	0.49	23.46%
Naïve Bayes	74.30%	0.45	25.70%

Auto ML

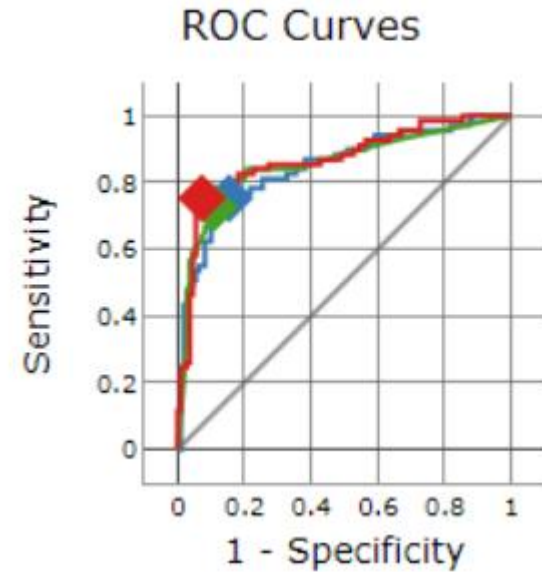
Model	Accuracy	Cohen's Kappa	Error (%)
XGBoost	85.47%	0.69	14.53%
Random Forest	82.12%	0.61	17.88%
Naïve Bayes	81.56%	0.61	18.44%

Evaluation

- **AutoML outperforms** manual modeling in every case, especially for XGBoost, where **Cohen's Kappa** rose from 0.52 to **0.69**, showing stronger agreement between predictions and true outcomes.
- **XGBoost (AutoML)** is the **best overall model**, with the highest **accuracy (85.47%)** and **lowest error (14.53%)**.
- Manual models show limited tuning — accuracies cluster around 74–78%, suggesting that **hyperparameter optimization** was minimal.

ROC Curve – Auto ML

- All three models show excellent performance (**AUC > 0.8**), confirming that the AutoML-tuned classifiers effectively differentiate between survivors and non-survivors in the Titanic dataset.
- XGBoost stands out again, combining high accuracy (85.47%), strong Kappa (0.69), and **the steepest ROC curve**, consistent with being the most robust model.





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